CHAPTER FOUR

DRAINAGE SYSTEMS AND WATER RESOURCE OF ETHIOPIA AND THE HORN

Objectives

At the end of this chapter you should be able to:

- Describe the major drainage systems in Ethiopia and the Horn,
- Examine the water resource potentials of Ethiopia,
- Understand economic potentials of the water sector in Ethiopia.

DRAINAGE SYSTEMS AND WATER RESOURCE OF ETHIOPIA AND THE HORN?...

Brain storming activity

- What is drainage and drainage system?
- What is mean by Ethiopia is '**Water Tower of Eastern Africa'?**
- What are the Major Drainage Systems of Ethiopia?



4.1. Introduction

- About 71% of the earth's total surface
- 97.5% is alkaline accumulated in seas and oceans.
- Only 2.5% is fresh water, of which;
  - ✓ nearly 68.7% is deposited in glaciers,
  - ✓ 30.1% in ground water,
  - ✓ 0.8% in permafrost and
  - ✓ 0.4% in surface waters.





Cont'd…

- Surface and ground waters are the most abundant and easily available fresh waters.
- However, fresh water is distributed unevenly throughout the world due to ***latitudinal***, ***climatic***, ***topographic*** factors.





Cont'd…

- The topographic setup of Ethiopia is complex blend of
  - massive highlands,
  - rugged terrain, and
  - low plains

and due to

- Diverse topographical setup,
- Relatively higher rainfall and
- Its nearness to equator made the country to have larger volume of ground and surface water.

- Consequently, 0.7 % of Ethiopia is covered by water bodies.

Definition of terms

- ***Drainage***:-flow of water through well-defined channel.
- **Drainage system** is made up of a principal river and its tributaries
- A ***river system*** begins at a place called the **source /headwater** and ends at a point called **mouth/outlet**.
- So, **a drainage system** is branched network of stream channels together with the adjacent land slopes they drain.

Cont…

- The ***drainage pattern*** of an area is the outcome of the geological processes, nature and structure of rocks, topography, slope, amount and the periodicity of the flow.
- A **drainage basin** is the topographic region from which a river and its tributaries collect both the surface runoff and subsurface flow.
- The general **patterns of major river basins** in Ethiopia are determined by topographical structures described below.

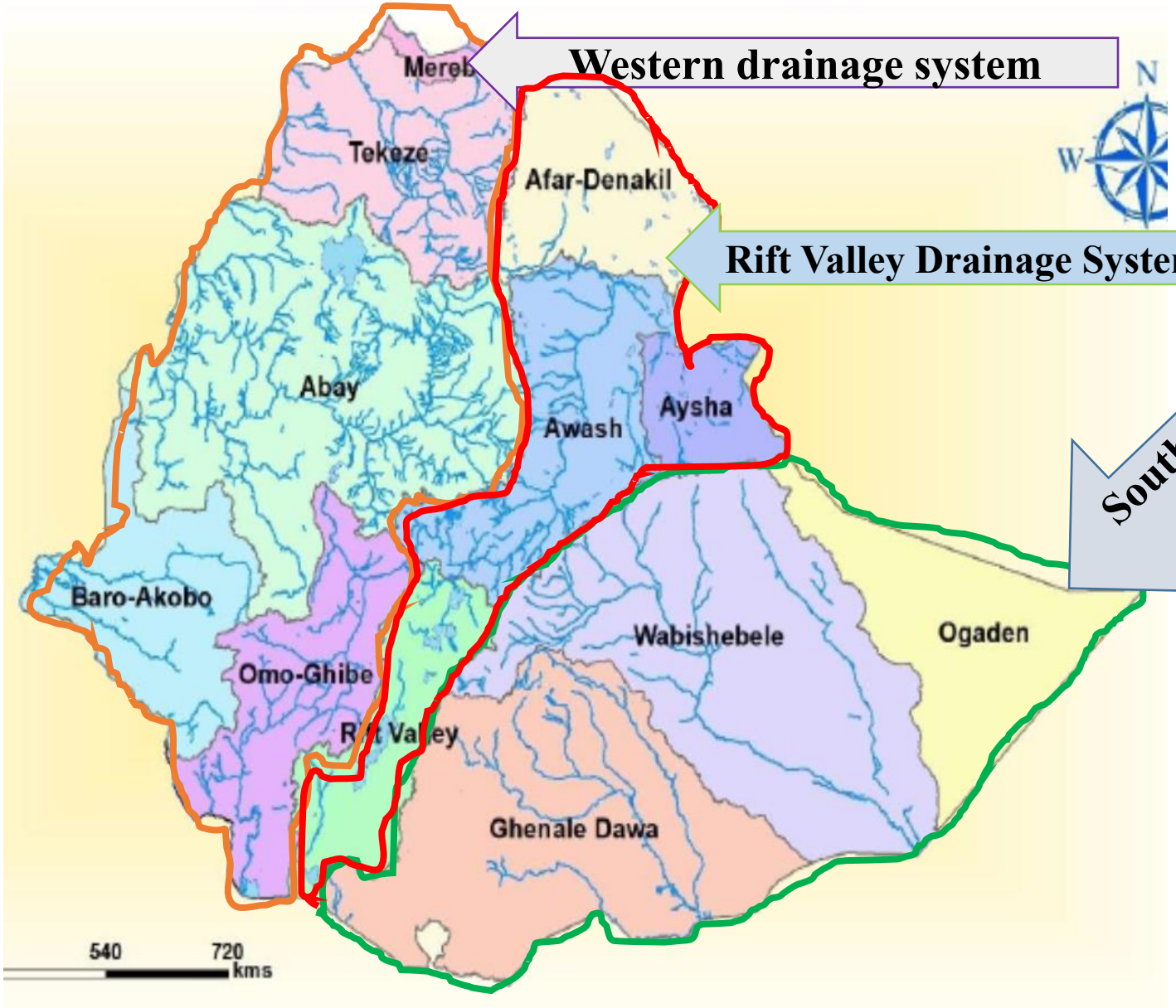
Cont'd…

a) The topography of the outward sloping of the **Western** and **South eastern plateaus**

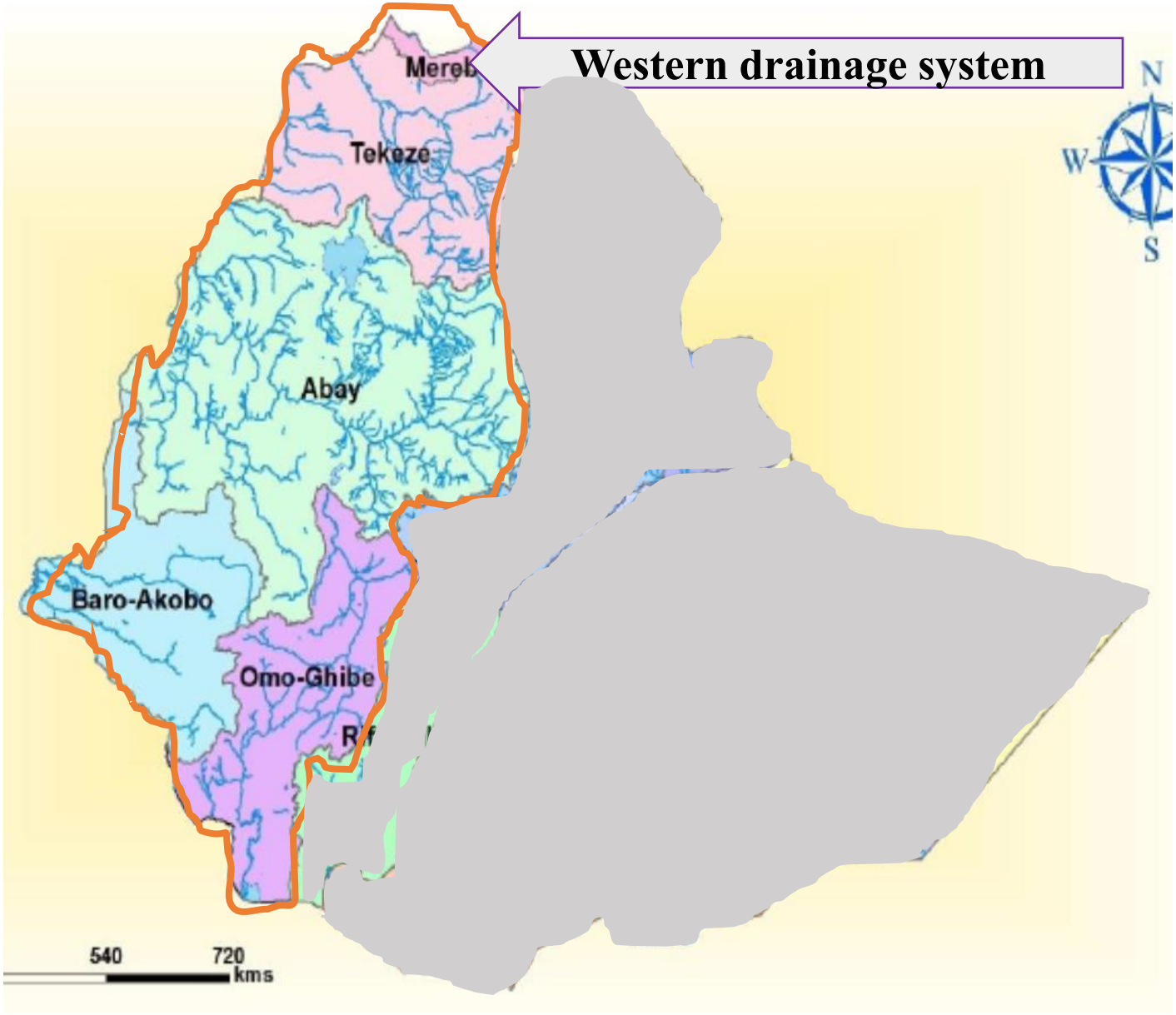
b) The structural formation of the **Rift Valley** with its inward-sloping escarpments resulting mainly in an inland drainage system.

c) Faults and **joints** that structurally influence part of the courses of many rivers.

Ethiopian drainage systems



(1) The Western Drainage Systems(WDS)



The Western Drainage Systems(WDS)…

- Are the largest of all DS draining 40 %of the total area of the country
- Carry 60 % of the annual water flow.
- Comprised of four major river basins:
  - Tekeze,
  - Abay,
  - Baro-Akobo and
  - Ghibe (Omo).
- Unlike Tekeze, Abay, and Baro-Akobo, which flow **westward**, the Ghibe (Omo) flows **southward.**

Abay River

- Is the **biggest** & **longest** river in the system
- Emerges from **Lake Tana?** & drains parts of the
- Is the **biggest river in terms of discharge** in Ethiopia.
- It joins the White Nile, which comes from Lake Victoria at the Sudanese Capital, Khartoum & drains the arid areas of both the Sudan and Egypt before it ends in the Mediterranean Sea.

Ghibe (Omo) River

- drain the wettest highlands in the south and southwestern Ethiopia.
- carry *6 percent* of the annual water flow
- drains an area of 79,000 km^2 with an estimated mean annual flow of 16.6 BMC.
- The ***Ghibe / Omo*** River finally empties in to the Chew-Bahir at the mouth of Lake Turkana (an elongated Rift Valley lake) thereby forming an inland drainage.

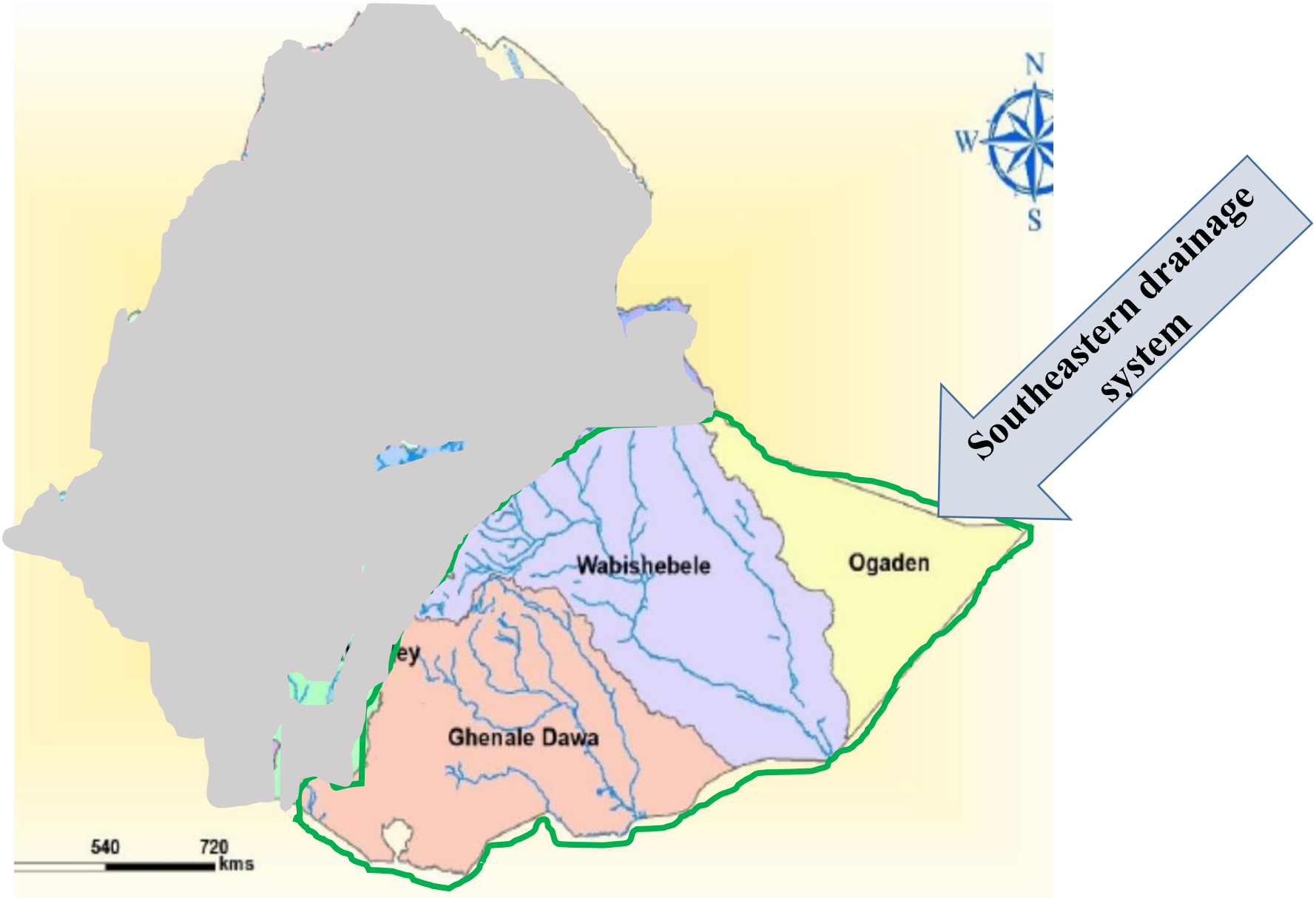
Tekeze River

- It drains parts of the **highlands in Tigray** & **North Central Massifs** & the **Anghereb- Tekkezze** lowlands.
- It is the **smallest** of the three in terms of amount of water flow in the region.
- **Tekeze river** is termed ***Atbara*** **in Sudan**, which is a tributary of the Nile.

Baro-Akobo River

- It descends from the **Southwestern highlands** & drains parts of higher & lower areas of the region; especially the **Gambella lowlands** take a huge advantage from it.
- Is the second **biggest river** in the region in terms of volume of water carried after the Abbay river.
- Along the Ethio-Sudan border it meets river **Akkobo** gets a new name in South Sudan i. e. **Sobat River**.

(2)The Southeastern Drainage Systems (SEDS)



(2) The Southeastern Drainage Systems (SEDS)…

- Entire physiographic region of southeastern part of Ethiopia is drained by this drainage systems.
    - Slopes: south-eastwards
- Major highlands of this basin include
    - plateaus of *Arsi*,
    - *Bale*,
    - Sidama and
    - Harerghe.
- **Wabeshebelle** and **Ghenale** carry 25 % of the annual flow of Ethiopia.
- ***Wabeshebelle*** is the largest (in catchment area in the region) and the longest river in Ethiopia.

- Despite its size, it fails to reach the Indian Ocean and disappears in the sands, just near the Juba River in Somali.
- *Ghenale (Juba) river* has fewer tributaries but carries more water than Wabeshebelle, reaches the Indian Ocean.

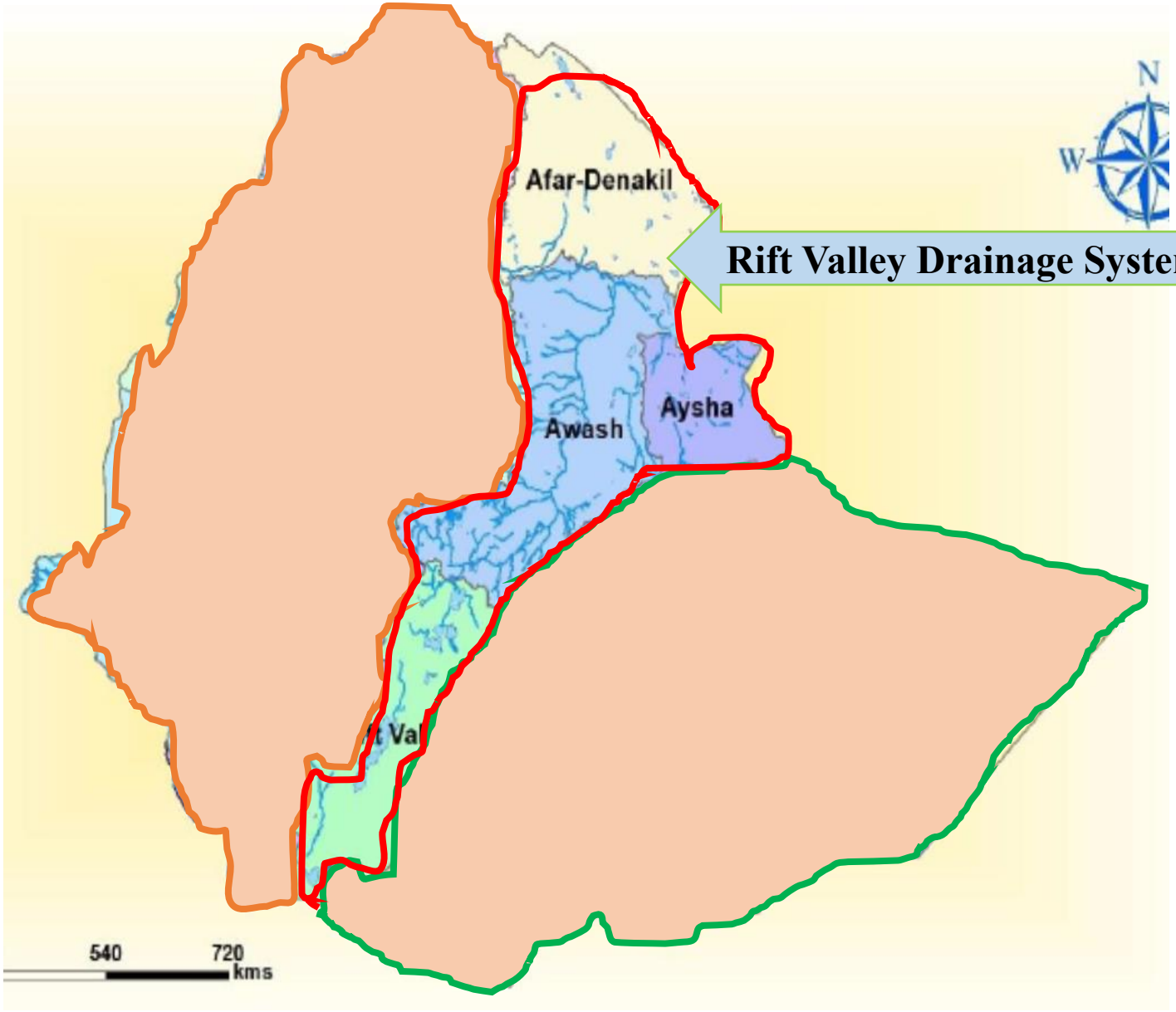
Wabishebelle River

- Emerges from the **Arssie-Bale massifs**
- Drains parents of the **Arssie-Bale highlands** (mainly the Hararghe plateaus) & the **Southeastern lowlands.**
- Flows through the dry lands of **Ogaden** & **Elkere** in Ethiopia & those in Somalia up until it dies in the sands along the coast of the Indian Ocean.
- It is the **longest** (2000kms) & **biggest river** (205,000 km^2) in Ethiopia.
- Despite its length & big areal size it carries **small amount of water** due to aridity & high evaporation.

Genale River

- Drains parts of the Southeastern highlands (mainly **Bale** & **Sidamo plateaus**) & lowlands of the southeast.
- Travels through more **humid areas**, thus carries **more water** than the Wabishebelle river.
- Along the Ethio-Somalia boreder at a place called **Dollo** it meets river Dawa & goes into Somalia by the name **Juba River.**
- By moving through the massive Borena lowlands & arid areas of Somalia, finally spills its water into the Indian Ocean.

(3) The Rift Valley Drainage System (RVDS)



(3) **The Rift Valley Drainage System (RVDS)…**

- The RVDS is an area characterized by
  - small amount of rainfall,
  - high evaporation and
  - small catchment area
- Awash river basin (most utilized river)
  - is the only major river of the RVDS
  - with the catchment area of 114,123 km2 and
  - 4.9 BCM average annual discharge.
- It originates from Shewa plateau (C. Ethiopia) covering some parts of;
  - Amhara,
  - Oromia,
  - Afar,
  - Somali,
  - Dire Dawa, and
  - Addis Ababa City Administration.

Cont'd…

- In this system, there is **no** one general flow direction as the streams flow in **all** directions.
- Awash flows in a **northeast** direction and finally ends up in **Lake Abe** along the Ethio-Djibouti border.
- ***The Afar drainage sub-basin*** has practically no stream flow and the only main surface waters in the basin are Lake Afrera and Asale, which were formed through tectonic activities.
- **The *Southern part of the Rift Valley sub-basin*** is characterized by a number of lakes and small streams and hence, described as 'lakes region'.

- There are *small streams* that supply water to lakes (e.g)
  - Meki and Katar Rivers flow into Ziway;
  - Bilate into Abaya; and
  - Segen into Chew Bahir
- Some of these lake are also **interconnected** (e.g.)
- Lakes Ziway and Langano drain into Lake Abijiata through the small streams of Bulbula and Horocolo respectively.

4.3. Water Resources: Rivers, Lakes and Sub-Surface Water

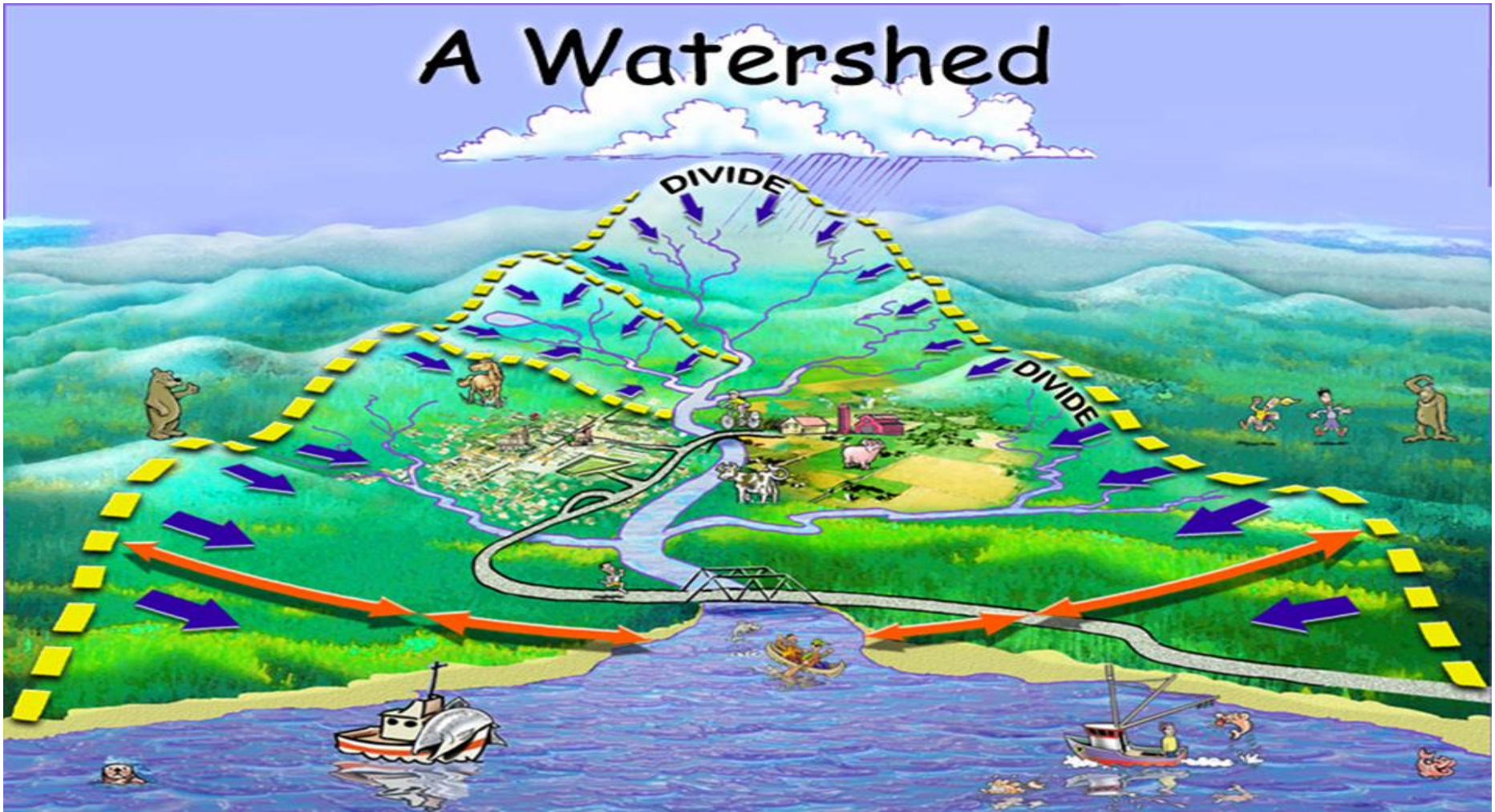
- Why do majority of Ethiopian rivers end at seas and oceans?

4.3.1. The Ethiopian Rivers

- Unlike many other African countries, Ethiopia is endowed with many rivers originating from highland areas and crossing the Ethiopian boundary.
- Altogether, Ethiopian rivers form *12 major watersheds* separating the Mediterranean Sea from the Indian Ocean drainage systems.

Cont'd…

- ***Watershed***
- Land area from which all water drains/flow to a destination such as a river, pond, stream, lake, or estuary.



4.3.2 Characteristics of Ethiopian Rivers

Ethiopian rivers have the following characteristics:

- ☞Almost all major rivers originate from the highlands with an elevation of >1500 masl.
- ☞**Trans-boundary** (rivers Abay, Tekkezze & Baro drain - Sudan & Egypt & finally ends at Mediterranean Sea).
- ☞**Extreme seasonal fluctuation(i.e**. high volume during wet season and low volume during the dry season)
- ☞**Rapids and waterfalls** along their course as they flow from highland to low land.
- ☞Have steep profiles as they flow on steep slopes.
- ☞Some of them serve as boundaries (both international and domestic administrative units).

4.3.2. **The Ethiopian Lakes**

- Ethiopia is rich in lakes and majority of lakes are located within the Rift Valley System.
- All of them are result of tectonic process-during Quaternary period of Cenozoic era.
- Lake Tana, -largest - & shallow depression formed through slower sinking and reservoir by lava flow b/n Gojjam and Gonder massifs.


Lake Tana

Cont…

- Ethiopia also has ‘crater lakes’ at and around
  - ***Bishoftu*,**
  - ***Wonchi*** (near Ambo),
  - *Hayk* (near Dessie) and
  - on top of *Mount Zikwala*.



Cont'd…

- ***Lake Ashenge*** (Tigray) is formed on a *tectonic basin*.
- *Man-made lakes*: such as Lakes Koka, Fincha and Melka Wakena, and many other lakes dammed following hydroelectric power generation projects.
- Main Ethiopian rift, Lake Abaya is the largest of all the lakes in the system.
- Southern tip of the Rift Valley, Chew Bahir is drained by Segan and Woito.
- Shala and Ziway are the deepest and the shallowest lakes in the central Ethiopian Rift respectively.

Cont'd…

Area and depth of some of Ethiopian Lakes

Lake	Area (km^2)	Max. Depth(m)	Lake	Area (km^2)	Max. Depth(m)
Tana	3600	9	Abijata	205	14
Abaya	1162	13.1	Awassa	129	10
Chamo	551	13	Ashenge	20	25
Ziway	442	8.95	Hayk	5	23
Shala	409	266*	Beseka	48.5	11
Koka	205	9			

Source: compiled from various sources

Cont'd…

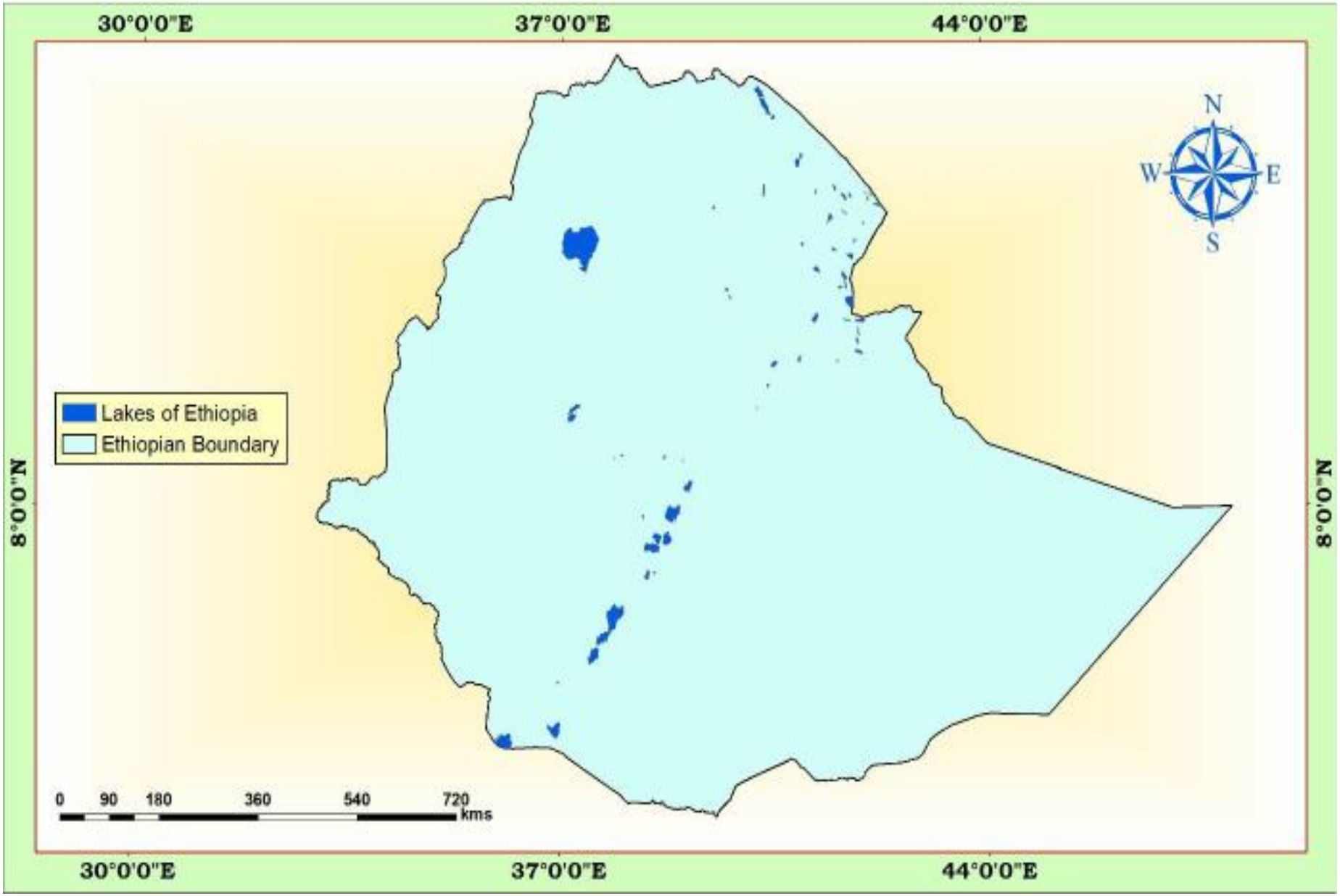


Figure 4.2. Lakes of Ethiopia Source: CSA shape file, 2012

CONT…

1. Based on Location:

- Highland &
- Rift Valley Lakes

I. Highland Lakes

- **Located in mountains** & **plateau regions** of the country.
- are either **crater** or **watershed** lakes.

❖**Watershed Lake-**created when rain water fills the valley or the lower places.

❖**Crater Lake (bowl-shaped geological formation at the top of a volcano)-** created when a volcano erupts and make a hole & hole is filled with water.

- Include lake Tana, which is the biggest &
- several small crater lakes in and around Bishoftu town, (Babo Gaya, Arsedi, Bishoftu, Hora, Kilote, & lake Arenguade).
- In addition to, Highland lakes also include Wonchi (north of Ambo),Hyke, Ashenge & Ziquala lakes.

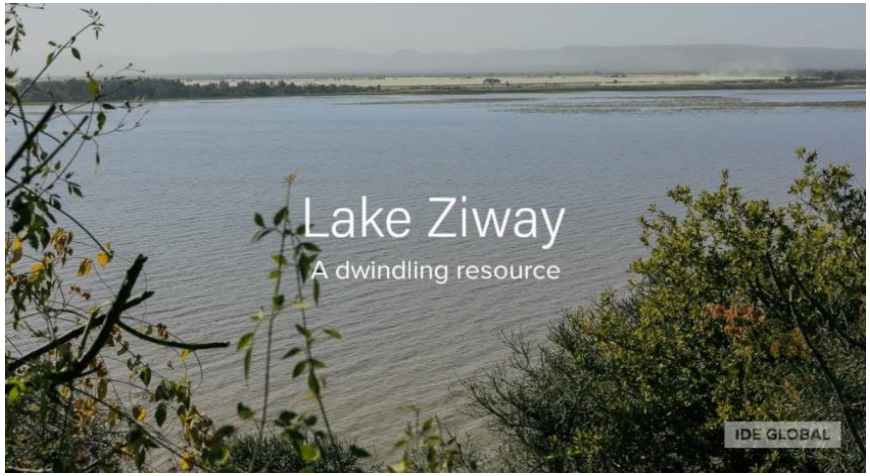
Con't…….

II. Rift Valley Lakes

- are found in the floor of the rift valley region of Ethiopia.

Included are lakes; Assale, Afdera, Abe, Beseka, Ziway, Langano, Abiatta, Shalla, Hawassa, Abaya, Chamo, Chewbahir.

While lake Abaya is the biggest of all the rift valley lakes by size, Shalla is the deepest of all Ethiopian lakes.





Con't…….

2. **Based on Creation**: Ethiopian lakes are broadly divided **into**

A. Natural &

B. Anthropogenic lakes.

- **Natural lakes** -evolved through geologic or climatic factors.
  - includes both the highland & rift valley lakes discussed above.
- **Anthropogenic ones-** emerged due to damming of rivers for various purposes-they are artificial.

Con't…….

- Koka, Fincha, Melkawakena, Tekkezze, Tanabeles, Gilgelghibe, Geferss, Legadadi, etc. are of Anthropogenic lakes in Ethiopia.

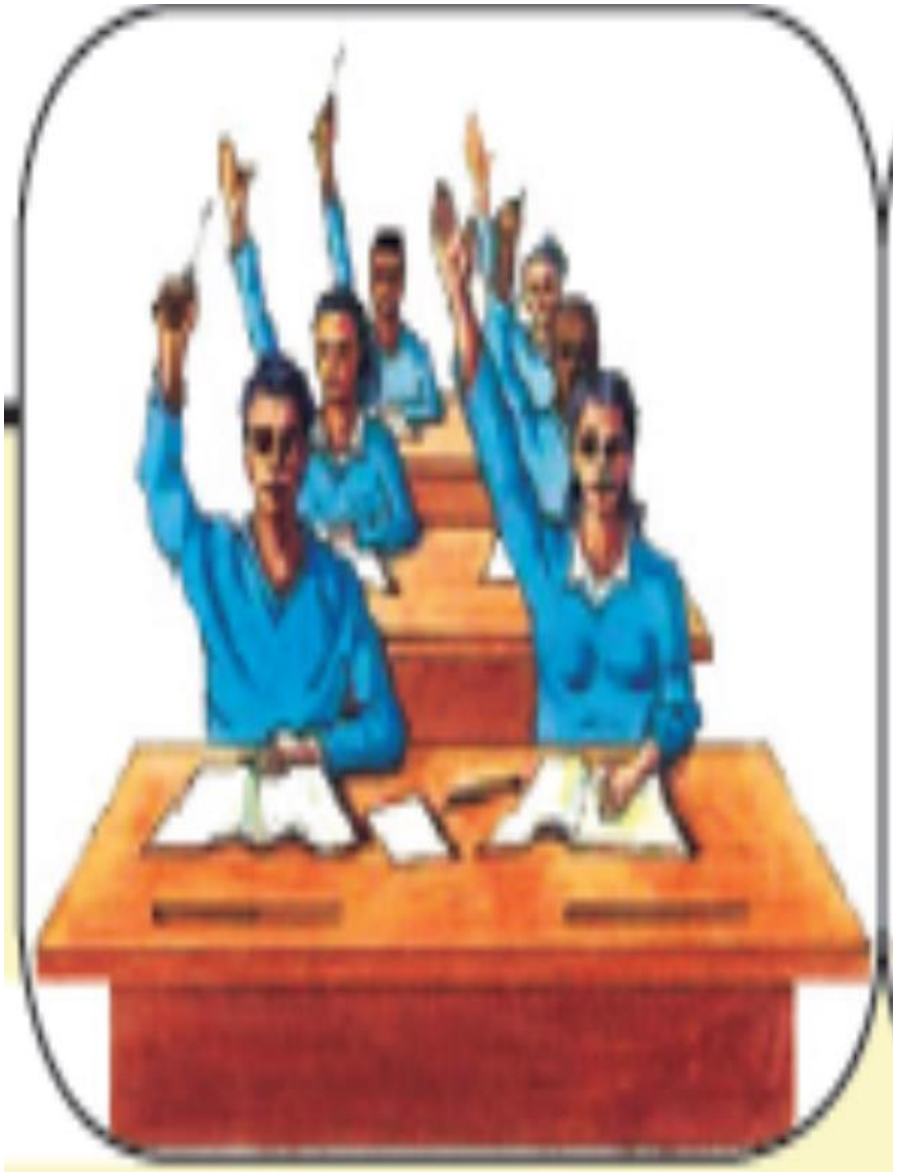




Subsurface (Ground) Water Resource of Ethiopia

Brainstorming questions:

1. What factors can determine the availability of ground water resources?
2. The potential of exploitable water in Ethiopia is huge, so why our agriculture is still rain fed?
3. What do you think that Ethiopian rivers are not conducive for navigation?
4. What factors made Ethiopia's water resources being underutilized?



Cont'd…

- Ethiopia has lower GW potential as compared to SW resources.
- However, there exists higher total exploitable GW potential.
- Based on existing scanty knowledge, GW potential is estimated to be 2.6 - 6.5 BMC (underestimated).
- Various studies shows that, Ethiopian potential of groundwater is believed to be b/n 12-30 BMC.

4.4. **Water Resources Potentials and Dev't in Ethiopia**

- Though Ethiopia benefited less from water resources, there are plenteous of opportunities that can transform the resource into societies collective social and economic needs.

❖Some of the '*potential development uses*' of water resource are:

- ☞Hydro-electric Potential
- ☞Irrigation and Transportation
- ☞Fishing and Recreation

Hydro-electric Potential

- Ethiopian rivers have a very high potential for generating electricity.
- Exploitable potential of HEP is estimated at about 45000 MW.
- Currently many HEP dams are operating and many others are under construction to realize Ethiopia's ambitious energy goals.
- *Grand Ethiopian Renaissance Dam (GERD)* is the country's largest dam under construction aiming to generate 6400 MW.

Cont'd…

- *Gilgel Gibe III* HP project has gone operational generating 1870 MW.
- Currently Ethiopia is administering 14 HE power plants constructed on Lake *Aba Samuel* (the first HE power generation plant installed in 1932),
  - Koka,
  - Tis Abay,
  - Awash,
  - Melka Wakena,
  - Sor,
  - Fincha,
  - Gibe/Omo,
  - Tana Beles and
  - Tekeze, generating close to 4000 MW of energy.

Irrigation and Transportation

- Uses of rivers for irrigation and transportation is constrained by the country's physiographic setups characterized by;
    - steep slopes,
    - rapids waterfalls,
    - narrow and deep valleys and
    - gorges in the high land areas.
- However, on lowlands, demand for irrigation is high and country's potential for irrigation is estimated to be 5.3 mll ha.
- The Baro-Akobo and Genale Dawa river systems have large irrigation potential compared to other basins.

Cont'd…

- Despite the **untapped** irrigation practice, >60% of the area under irrigation so far is located in Rift Valley Drainage System.
- Most of Ethiopian rivers are not suitable for transportation and ***Baro river*** is the only navigable river at its lower course.
- Comparatively, Ethiopian lakes are much suitable for transportation than rivers
    - Lake Tana and
    - Abaya are relatively the most used for transportation.

Fishing and Recreation

- Ethiopian lakes are rich in fish and The present annual production of fish is estimated to be 31.5 thousand tons.
- The exploitable potential is however, by far greater than the current production. Exploitable fish potential is vary from lake to lake.
- Currently, Lake Tana; the leading potential of fish exploitation estimated to be b/n 8,000-10,000 t/yr and followed by
- Lake Chamo having Fish production estimated at 4,500 t/yr.

Cont'd…

- However, >60% of fish supplies are from RV lakes.
- Some of the lakes are currently threatened by
  - sedimentation,
  - invasive species (water hyacinth),
  - over exploitation and
  - expansion of investments around lakes.
- Apart from being the source of fish, many Ethiopian lakes also serve for
  - recreational and
  - tourist attractions site due to:
    - ☞The presence of variety of fish, birds and other aquatic life forms
    - ☞the scenic beauty of the lakes
    - ☞the hot springs around the lakes
    - ☞the spectacular river gorges and the most impressive waterfalls, …

Main points of the lesson

- Drainage system
  - Western, Southeastern and Rift Valley
- Ethiopian Rivers
  - Characteristics
- Ethiopian Lakes
  - Types of lakes based on different criteria's
- Sub-surface Water resource of Ethiopia
- Water resource potentials and development
  - Hydro –power, fishing and recreation, irrigation and transportation, industrial and domestic uses